

ICPSR/Retraction Watch, First Look

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DOIs available?

```
# What proportion of entries in each table have DOIs (ICPSR, Retraction Watch)
c(length(icpsr$DOI[which(!is.na(icpsr$DOI))]) / length(icpsr$DOI),
  length(retractions$OriginalPaperDOI[which(!is.na(retractions$OriginalPaperDOI))]) /
  length(retractions$OriginalPaperDOI))
```

```
## [1] 0.4917015 0.9444636
```

```
# Duplicate DOIs in retractions?
```

```
retractions |>
  drop_na(OriginalPaperDOI) |>
  add_count(OriginalPaperDOI) |>
  filter(n > 1) |>
  distinct() |>
  select("Record.ID", "Title", "OriginalPaperDOI", "n") |>
  nrow()
```

```
## [1] 10821
```

```
# Retraction Watch database contains 'duplicate' DOIs - if a paper
# was corrected then retracted, there are separate entries for the
# correction and the retraction
```

```
# pivot_wider before merging?
```

```
# How many entries for each type of retraction reason?
```

```
retractions |>
  drop_na(RetractionNature) |>
  group_by(RetractionNature) |>
  summarize(n())
```

```
## # A tibble: 4 x 2
##   RetractionNature      `n()`
##   <chr>              <int>
## 1 Correction          1362
## 2 Expression of concern 3440
## 3 Reinstatement        158
## 4 Retraction          61681
```

```
# Duplicate DOIs in ICPSR?
```

```
icpsr |>
  drop_na(DOI) |>
```

```
add_count(DOI) |>
filter(n > 1) |>
distinct() |>
select("Reference.ID", "Title", "DOI", "n") |>
nrow()
```

```
## [1] 0
```

```
# Doesn't look like it! Thank you data librarians :)
```

RIS Types

```
# Article type found in 'RIS.Type' column, not 'Article.Type'

# For entries without DOIs in ICPSR, what type of materials are they?

# 'RIS.Type' available for what proportion of icpsr entries?
icpsr |>
  drop_na(RIS.Type) |>
  select("Reference.ID", "Title", "DOI", "RIS.Type") |>
  nrow() / nrow(icpsr)
```

```
## [1] 1

# 'RIS.Type' available for what proportion of icpsr entries
# that have a DOI?
icpsr |>
  drop_na(DOI) |>
  drop_na(RIS.Type) |>
  select("Reference.ID", "DOI", "Type.of.Work") |>
  nrow() / nrow(filter(icpsr, is.na(DOI)))
```

```
## [1] 0.9673478

# ... that don't have a DOI?
icpsr |>
  filter(is.na(DOI)) |>
  drop_na(RIS.Type) |>
  select("Reference.ID", "Title", "DOI", "Type.of.Work") |>
  nrow() / nrow(filter(icpsr, is.na(DOI)))
```

```
## [1] 1

# 'Type.of.Work' appears to be available for < 10% of entries -
# around 5.4% of entries with a DOI and 18.5% of those w/o a DOI

# TODO: fix NA values - diff NA values associated w diff columns

# How many entries associated with each type of work?
icpsr |>
  drop_na(RIS.Type) |>
  filter(RIS.Type != "(unknown)") |>
  group_by(RIS.Type) |>
  summarize(count = n()) |>
  arrange(desc(count))
```

```
## # A tibble: 12 x 2
##   RIS.Type count
##   <chr>    <int>
## 1 JOUR      70409
## 2 RPRT      15470
## 3 CONF      11533
## 4 THES      10349
## 5 CHAP       5148
## 6 BOOK       2809
## 7 GEN        1387
```

```
## 8 MANSCPT      686
## 9 NEWS          575
## 10 ELEC         548
## 11 MGZN         340
## 12 ADVS         225
```

Needs to be standardized .. capitalization differences interfering

'ArticleType' unavailable for what proportion of RW entries?

```
retractions |>
  select(Record.ID, ArticleType) |>
  drop_na(ArticleType) |>
  nrow()
```

```
## [1] 66641
```

ArticleType available for all RW entries, it looks like!

How many entries associated with each type of work in RW?

```
retractions |>
  select(Record.ID, ArticleType) |>
  drop_na(ArticleType) |>
  unnest(c(ArticleType)) |>
  group_by(ArticleType) |>
  summarize(count = n()) |>
  arrange(desc(count))
```

```
## # A tibble: 26 x 2
##   ArticleType      count
##   <chr>          <int>
## 1 Research Article 45084
## 2 Conference Abstract/Paper 13280
## 3 Clinical Study  2970
## 4 Review Article  2581
## 5 Meta-Analysis   875
## 6 Case Report     868
## 7 Book Chapter/Reference Work 576
## 8 Article in Press 519
## 9 Letter          517
## 10 Commentary/Editorial 443
## # i 16 more rows
```

These overlap - ArticleType is a list-type column

Publishers

```
# Publication name available for what proportion of entries?
```

```
# Secondary.Title includes publication name if entry is  
# a journal/newspaper
```

```
icpsr |>  
  select(Reference.ID, Secondary.Title) |>  
  drop_na(Secondary.Title) |>  
  nrow() / nrow(icpsr)
```

```
## [1] 0.8259025
```

```
# Now looking at Publisher - contains info on gov't  
# bureaus etc responsible for the publication  
# Approx equivalent to 'Institution' in RW data?
```

```
icpsr |>  
  select(Reference.ID, Publisher) |>  
  drop_na(Publisher) |>  
  nrow() / nrow(icpsr)
```

```
## [1] 0.3594607
```

```
# Around 36% of icpsr entries list the publisher
```

```
# What about those entries without DOI?
```

```
# Secondary.Title
```

```
icpsr |>  
  filter(is.na(DOI)) |>  
  select(Reference.ID, Secondary.Title) |>  
  drop_na(Secondary.Title) |>  
  nrow() / nrow(filter(icpsr, is.na(DOI)))
```

```
## [1] 0.6794059
```

```
# Publisher
```

```
icpsr |>  
  filter(is.na(DOI)) |>  
  select(Reference.ID, Publisher) |>  
  drop_na(Publisher) |>  
  nrow() / nrow(filter(icpsr, is.na(DOI)))
```

```
## [1] 0.6396239
```

```
# Around 64% of entries without a DOI have a publisher listed
```

```
# What publications appear frequently in ISPCR?
```

```
# Secondary.Title
```

```
icpsr |>  
  select(Reference.ID, Secondary.Title) |>  
  drop_na(Secondary.Title) |>  
  group_by(Secondary.Title) |>  
  summarize(count = n()) |>  
  arrange(desc(count))
```

```
## # A tibble: 19,705 x 2
##   Secondary.Title                                count
##   <chr>                                           <int>
## 1 ProQuest Dissertations and Theses              1021
## 2 Journal of Marriage and Family                  971
## 3 American Journal of Public Health              842
## 4 Journals of Gerontology, Series B: Psychological Sciences and Social S~ 649
## 5 Social Forces                                  635
## 6 Social Science Quarterly                      627
## 7 annual meeting of the American Political Science Association          618
## 8 American Sociological Review                  586
## 9 American Journal of Political Science          583
## 10 American Political Science Review             556
## # i 19,695 more rows
```

```
# Publisher
```

```
icpsr |>
  select(Reference.ID, Publisher) |>
  drop_na(Publisher) |>
  group_by(Publisher) |>
  summarize(count = n()) |>
  arrange(desc(count))
```

```
## # A tibble: 6,988 x 2
##   Publisher                                count
##   <chr>                                   <int>
## 1 American Society of Criminology           965
## 2 United States Department of Justice, National Institute of Justice      879
## 3 U.S. Department of Health and Human Services, Administration for Child~ 631
## 4 National Bureau of Economic Research       603
## 5 United States Department of Justice, Bureau of Justice Statistics         600
## 6 American Sociological Association          573
## 7 United States Department of Education, Office of Educational Research ~ 448
## 8 University of Michigan                    428
## 9 Substance Abuse and Mental Health Services Administration               420
## 10 University of North Carolina at Chapel Hill, Carolina Population Center 419
## # i 6,978 more rows
```

```
# Publication name available for what proportion of entries in RW?
```

```
# Secondary.Title includes publication name if entry is
# a journal/newspaper
```

```
retractions |>
  select(Record.ID, Journal) |>
  drop_na(Journal) |>
  nrow() / nrow(retractions)
```

```
## [1] 1
```

```
# All entries list a 'Journal' - column just holds the source
# of the article (incl. journals, books, serials, etc)
```

```
# What journals are most common in RW?
```

```
retractions |>
  select(Record.ID, Journal) |>
```

```
drop_na(Journal) |>
unnest(c(Journal)) |>
group_by(Journal) |>
summarize(count = n()) |>
arrange(desc(count))
```

```
## # A tibble: 8,496 x 2
##   Journal                                count
##   <chr>                                <int>
## 1 Journal of Intelligent & Fuzzy Systems      1566
## 2 2011 International Conference on E-Business and E-Government (ICEE)    1280
## 3 PLoS One                                  1224
## 4 2011 5th International Conference on Bioinformatics and Biomedical Eng~ 1084
## 5 Journal of Healthcare Engineering          1074
## 6 Computational and Mathematical Methods in Medicine    1067
## 7 Computational Intelligence and Neuroscience    1028
## 8 BioMed Research International              953
## 9 Security and Communication Networks           949
## 10 Journal of Physics: Conference Series        878
## # i 8,486 more rows
```

Merging

```
# Initial brief attempt at a merge
```

```
merged <- inner_join(
  x = (icpsr %>%
    filter(!is.na(DOI)) %>%
    select(Reference.ID, Title, DOI) %>%
    unnest(c(DOI))),
  y = (retractions %>%
    filter(!is.na(OriginalPaperDOI)) %>%
    select(Record.ID, Title, OriginalPaperDOI, RetractionDate, RetractionNature, Reason) %>%
    unnest(c(OriginalPaperDOI))),
  by = join_by(DOI == OriginalPaperDOI)
)

print(merged)
```

```
## # A tibble: 9 x 8
##   Reference.ID Title.x   DOI   Record.ID Title.y RetractionDate RetractionNature
##   <int> <chr>    <chr> <list>    <list>    <list>         <chr>
## 1    11313 Worlds A~ 10.1~ <chr [1]> <chr>    <chr [1]>    Retraction
## 2    15930 Nonmetro~ 10.1~ <chr [1]> <chr>    <chr [1]>    Retraction
## 3    29220 School s~ 10.1~ <chr [1]> <chr>    <chr [1]>    Retraction
## 4     87821 Suicide ~ 10.1~ <chr [1]> <chr>    <chr [1]>    Retraction
## 5    140614 Electron~ 10.1~ <chr [1]> <chr>    <chr [1]>    Correction
## 6    140614 Electron~ 10.1~ <chr [1]> <chr>    <chr [1]>    Retraction
## 7    142038 The verd~ 10.1~ <chr [1]> <chr>    <chr [1]>    Retraction
## 8    144102 Chronic ~ 10.1~ <chr [1]> <chr>    <chr [1]>    Correction
## 9    155116 Tuning i~ 10.1~ <chr [1]> <chr>    <chr [1]>    Expression of c~
## # i 1 more variable: Reason <list>
```

```
# Is 9 observations around what we'd expect??  
# Slightly concerning might be a data cleaning problem  
  
# remove duplicate row from merged  
merged = merged[c(1:4, 6:9),]
```